

Signal and System Experiment Tutorial

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Instrument Introduction

The hardware is primarily based on a virtual instrumentation platform. The virtual instrumentation platform selected for this book is NIELVIS II+, which integrates common electronic measuring instruments such as function generator, a dual-channel digital oscilloscope, a multimeter, and DC power supplies. The model of Signal and System experiment board is EMONA SIGEx (referred to as SIGEx hereinafter), which is placed on the virtual instrument platform and stably connected through a dedicated interface. The experiment board contains multiple independent unit circuit modules. Each module implements a simple function, and combining several modules enables the completion of numerous different experiments. Modules are connected using wires. The connection circuit diagram may refer to the corresponding figure in this book or be freely configured according to specific needs. The NIELVIS II+/SIGEx kit is shown in Figure 0.1.

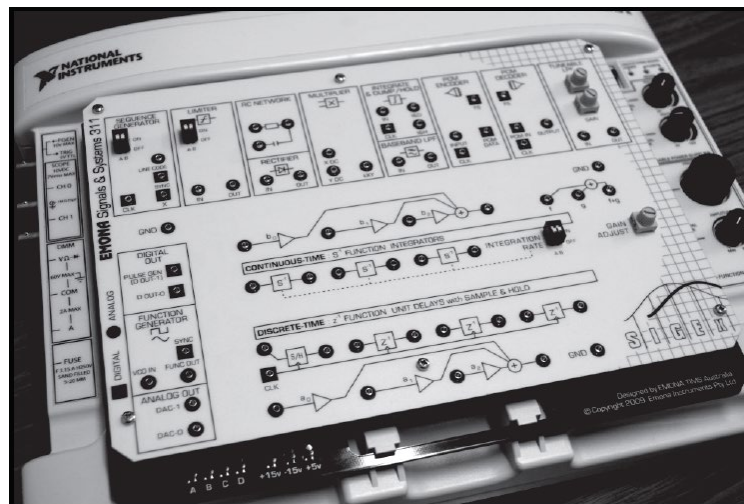


Figure 0.1 NI ELVIS II+/SIGEx Kit

The following is a brief introduction to each module.

1. SEQUENCE GENERATOR

The Sequence Generator provides a periodic data source, outputting 5V standard bipolar logic levels. Different data patterns can be selected via the switch in the upper-left corner, and the state of this switch can be displayed within the experimental program. This module is clocked by a logic level signal (CLK). It outputs one cycle synchronization (SYNC) pulse per frame. Typically, this logic level clock is provided by either a Pulse Generator, a Function Generator or the Sync Output (FUNCTION GENERATOR/SYNC).

2. LIMITER

The Limiter uses switches to select gain levels, amplifying the input signal to a fixed value to

produce an output signal with limited amplitude.

3. RC NETWORK

The RC Network contains individual resistor and capacitor components. These components can be used independently by wiring them, or combined to form RC circuits, typically used as low-pass filters. All components are independent, and attention must be paid to connecting Ground (GND) during use.

4. RECTIFIER

The Rectifier provides half-wave rectification of the input signal. It consists of a non-ideal diode with a forward voltage drop and is generally intended for the use of sinusoidal signals.

5. MULTIPLIER

The Multiplier provides the multiplication function for two analog input signals. With a k factor approximately equal to 1, the output kXY equals the signal resulting from the multiplication of input signals X and Y .

6. INTEGRATE & DUMP/HOLD

This module can implement steep fall in integrate and integrate-and-hold functions. A clock input signal is required, operating only over one cycle.

7. BASEBAND LPF

The baseband low-pass filter is a fourth-order Butterworth low-pass filter, which can be used either as a system under test or as a general filter.

8. PCM ENCODER

This module implements PCM encoding for a single channel of analog signal. It outputs an 8-bit frame, which includes a periodic frame synchronization pulse. It can handle both DC signals and sinusoidal signals. Its maximum sampling rate is 2.5 KS/s, making it suitable for signals with frequencies below 1.25 kHz.

9. PCM DECODER

This module implements PCM decoding for 8-bit PCM data streams. The Frame Sync (FS) signal and the sampling frequency clock must be correctly connected.

10. TUNEABLE LPF

This is an 8th-order elliptic low-pass filter with adjustable center frequency and gain. It operates on analog signals and TTL-level digital signals. The filter has no anti-aliasing measures at its input; therefore, the bandwidth of the input signal must be explicitly defined during work.

11. INTEGRATORS

The Integrators module contains three independent simple integrator circuits. A switch on the right allows adjustment of the integration rate. These integrators are designed for continuous time-domain integration. The switch settings and integration rates can be displayed within the experiment program.

12. UNIT DELAYS with SAMPLE & HOLD

The Sample/Hold is an analog sampling circuit that can sample the input signal based on the clock signal and hold the sampled value for one clock. The unit delay causes the value of the analog signal to be output with a delay of one clock cycle. Its clock and sampler/holder are the same input, and the four independent delay units share the same clock signal.

13. ADDERS and GAIN ADJUST KNOB

This module contains three adders. Two adders are three-input with adjustable coefficients, while the third is a two-input adder. For the three-input adders (a and b), the gain coefficient for each input channel can be adjusted via the experiment program, allowing the construction of custom experimental systems as needed. The two-input adder has a fixed gain of 1, implementing a simple addition function. The value of the gain adjustment knob can be manually adjusted and is displayed within the experiment program.

14. DIGITAL OUT

This module is a built-in pulse generator. Its frequency and duty cycle can be controlled via the experiment program, and it is commonly used as a digital clock source.

15. FUNCTION GENERATOR

This module corresponds to the built-in multi-function signal generator within the virtual instrument platform. It can provide various types of signals with controllable amplitude and frequency. Parameters for the function generator signal can be set via the virtual instrument platform's control software or through the experiment program.

16. ANALOG OUT

This module generates two channels of analog output via digital to analog converter (DAC). These outputs are controlled by the experiment program and can be changed according to user requirements to produce arbitrary periodic waveforms.

17. Experiment Program

The control software of the signal and system experiment board can call the control ports of virtual instruments and directly control hardware devices such as oscilloscopes, signal generators, and pulse sequence generators on the virtual instrument platform in the program. The software contains multiple tabs (Lab) resources, and different experimental projects can select different tabs for experiments. The front panel of EMONA SIGEx software is shown in Figure 0.2.

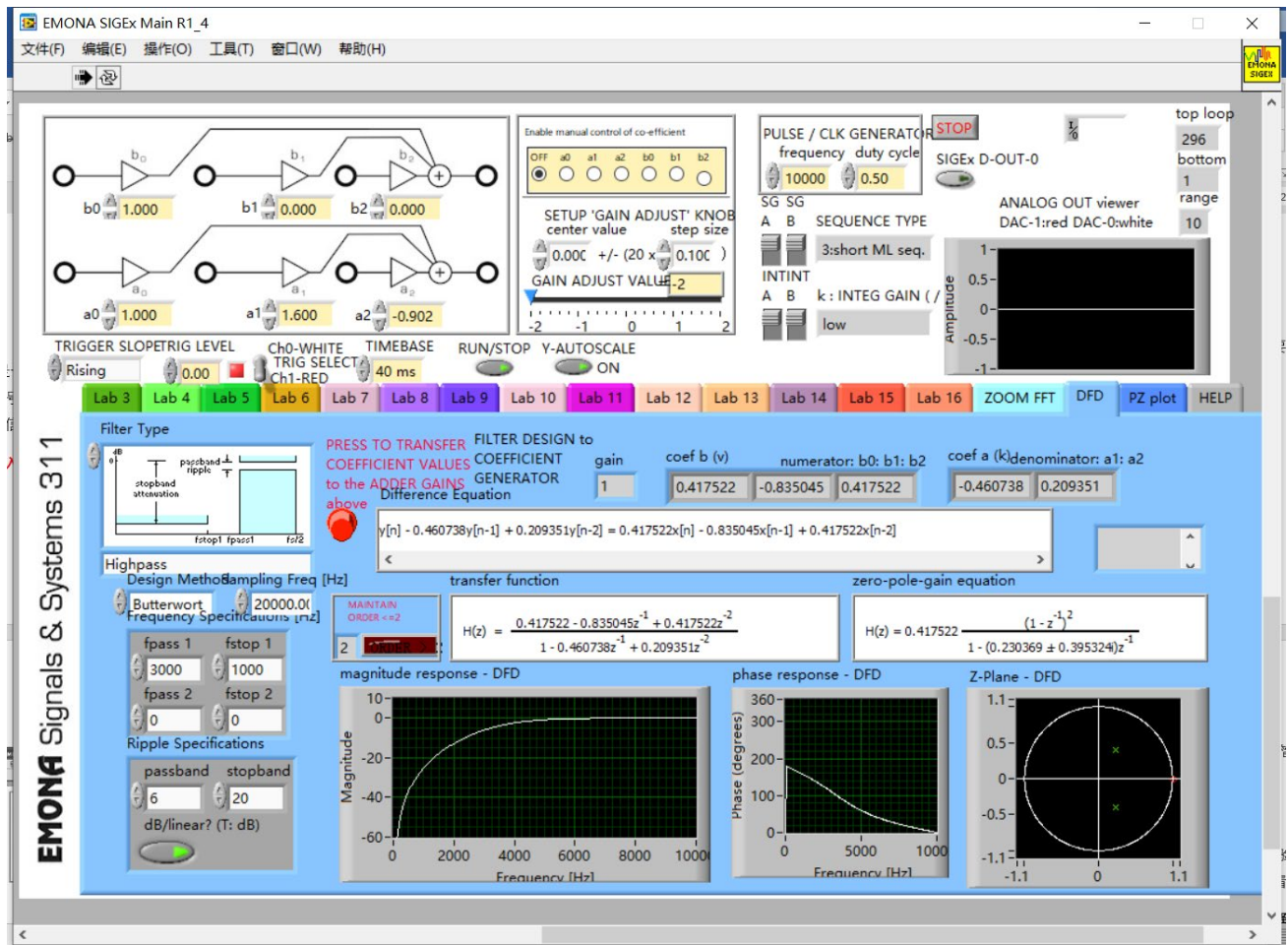


Figure 0.2 Front panel of EMONA SIGEx

Experiment 1 Characteristics of Signals

I. Purpose of the Experiment

- (1) Learn the usage of the virtual instrumentation platform and the Signal and System experiment board.
- (2) Understand the impact of baseband low-pass filter (BLPF), tunable low-pass filter (TLPF), and RC circuits on signals.
- (3) Understand the principles of limiters.
- (4) Understand the law of how pulse sequence rates are affected by system characteristics and the method of calculating the maximum slew rate.

II. Pre-lab Requirements

- (1) Review basic circuit knowledge (RC circuits) involved in the experiment.
- (2) Preview the usage and precautions of the instruments used in the experiment.

III. Experimental Instruments and Components

Table 1.1 List of Experimental Instruments and Components

Name	Quantity	Model/Type
Virtual Instrumentation Platform	1 set	NI ELVIS II+
Signal & System Experiment Board	1 set	EMONA SIGEX
Computer	1 unit	\
BNC-terminated Cables	3	2 mm
Connection Wires	several	2 mm

IV. Experimental Principles

1. Characteristics of Pulse Signals

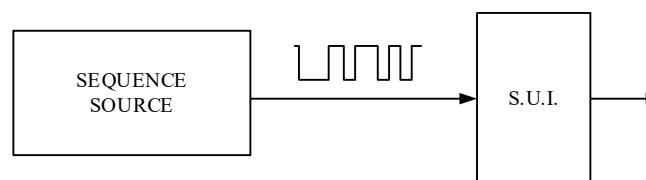


Figure 1.1 Digital Pulse Sequence passes through System

Digital pulse waves are typically transmitted through a system under investigation (SUI), where they can be amplified, filtered, etc. This experiment explores signal characteristics when transmitted through a channel with the limited slew rate using simple analog filters.

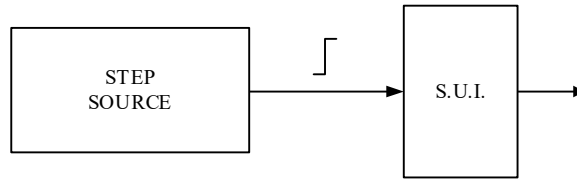


Figure 1.2 Step signals passes through system

Step signals are fundamental components of pulse signals. The transition time, transition form, oscillation duration, and amplitude collectively determine the properties of the output step signal.

2. Characteristics of Sinusoidal Signals

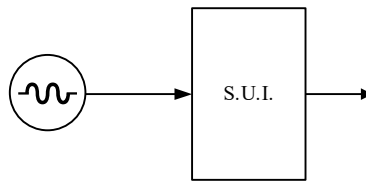


Figure 1.3 Sinusoidal signals pass through system

This experiment explores the characteristics of signals transmitted through a slew-rate-limited system.

3. Limiting

Voltage limiting is a common effect in circuits. A typical example is in amplifier circuits where the signal amplitude is clipped when it exceeds the DC power supply voltage limits.

V. Experimental Procedure

1. Preparation

Before conducting experimental operations, please first check the instruments according to the pre-experiment requirements. The inspection includes:

- (1) Ensuring the virtual instrumentation platform hardware functions work normally.

Verify that the platform's power is turned on (indicated by the illuminated status light at the top-right of the instrument) and that the Signal and System experiment board is correctly installed (indicated by the illuminated status light at the bottom-left of the instrument), as shown in Figure 1.4.

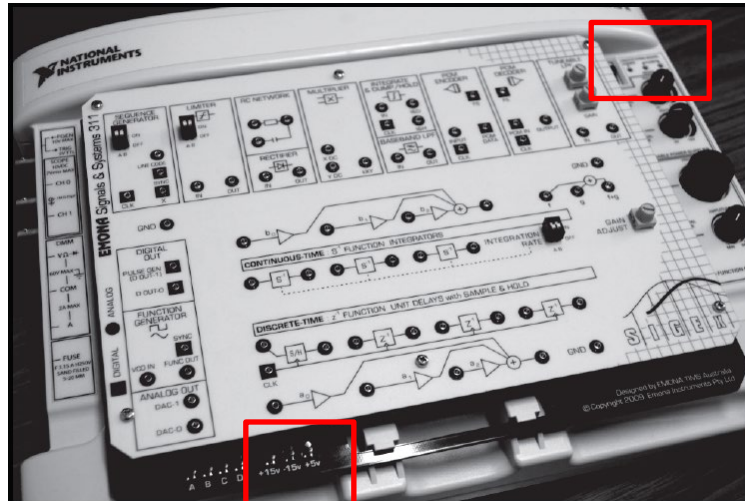


Figure 1.4 NI ELVIS II/SIGEx Kit

(2) Launching the virtual instrument control interface.

When a virtual instrumentation platform is connected to the computer, the instrument startup panel should run automatically. If it does not launch automatically, please click the NI ELVISmx Instrument Launcher icon to open the software, as shown in Figure 1.5.

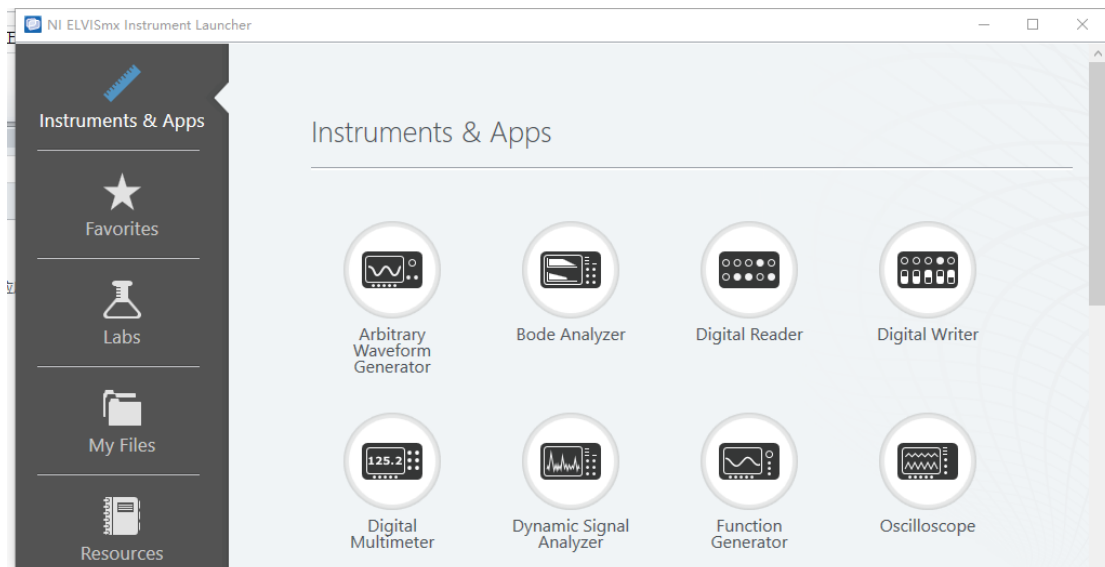


Figure 1.5 NI ELVIS II instrument launcher panel

This software serves as the control interface of the virtual instrumentation platform hardware. It provides virtual panels to operate common instruments such as the oscilloscope, function generator, and digital multimeter. Signal transmission and acquisition occur physically, while the instrument control interface is virtualized.

(3) Opening the Signal and System Experiment Board control software (SIGEx).

Launch the SIGEx R1-4 main program. When the program runs successfully, the red LED at the bottom-left of the virtual instrumentation platform hardware will illuminate, and the main program interface will appear as shown in Figure 1.6. If the program encounters errors, please first verify hardware connections; then click the "STOP" button to halt the program, followed by the "Run" button

to restart program execution.

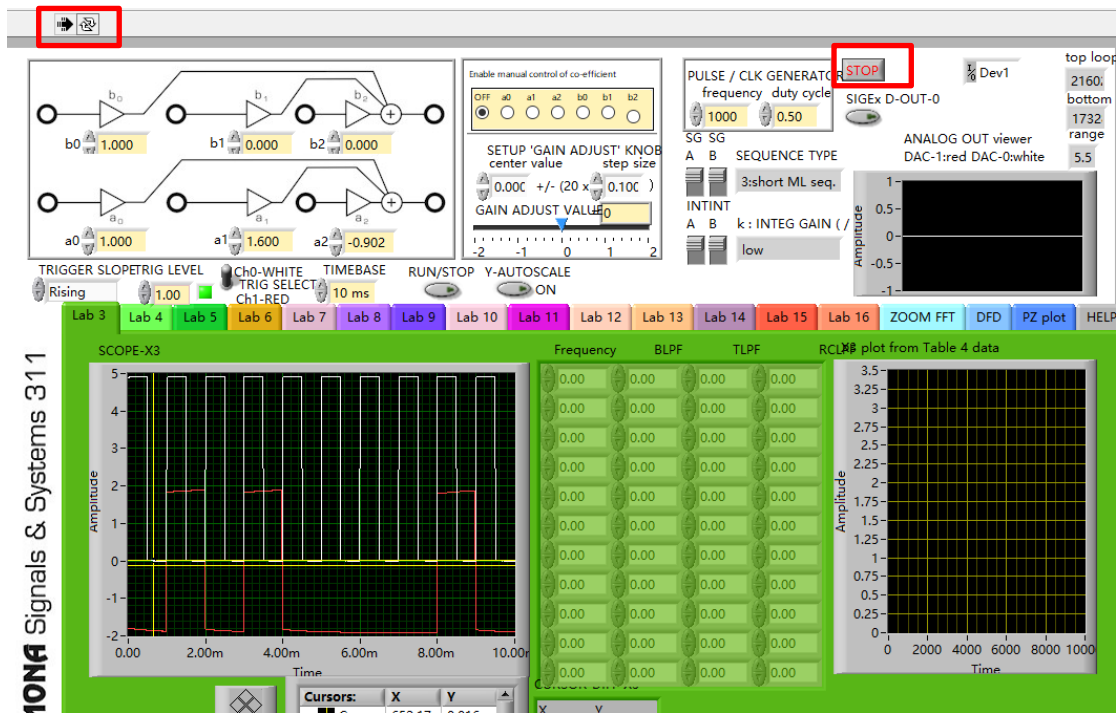


Figure 1.6 SIGEx Software Interface

Only after completing the above steps may the experiment commence.

2. Characteristics of Pulse Signals

(1) Sequence Generator.

Open the SIGEx program, select Lab3 tab, and use cables and wires to connect the circuit according to Figure 1.7. Set the parameters as follows:

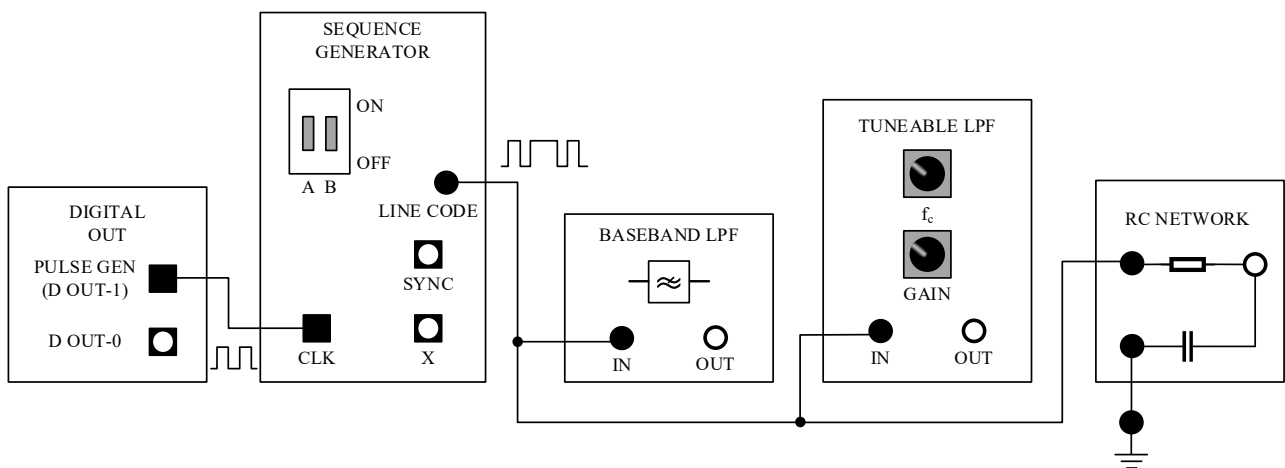


Figure 1.7 Wiring diagram for sequence wave passing through filters

①Pulse Generator: Frequency = 1000 Hz, Duty Cycle = 50%.

②Sequence Generator: Set both DIP switches to the up position to generate a short sequence signal (short ML seq.).

③Oscilloscope: Set the timebase to 10 ms/div, select CH0 channel, use the rising edge as the trigger, and set the trigger level to 1 V.

Question 1.1

Connect CH0 channel to the Pulse Generator output (PULSEGEN) and CH1 channel to the Sequence Generator output (LINE CODE). Observe the sequence generated by the Sequence Generator and determine the minimum and maximum intervals between two adjacent pulses ()

- A. 0.1-0.4 ms, 10-9 ms
- B. 0.1-0.4 ms, 4-5 ms
- C. 1-2 ms, 4-5 ms
- D. 1-2 ms, 10-9 ms

(2) Baseband Low-Pass Filter and Tunable Low-Pass Filter.

Connect CH0 channel to the output of the Baseband Low-Pass Filter (BASEBAND LPF, BLPF) and CH1 channel to the output of the Tunable Low-Pass Filter (TUNEABLE LPF, TLPF). Use "RUN/STOP" and "Y-AUTOSCALE" buttons to pause or stabilize the measured signals for easier reading.

On the experiment board, adjust the center frequency (f_c) knob of TLPF clock-wise to the end, and adjust the gain (GAIN) knob of TLPF where TLPF's amplitude matches BLPF's amplitude as much as possible, as shown in Figure 1.8.



Figure 1.8 Diagram of BLPF and TLPF outputs

Question 1.2

What changes occur to the signal after passing through the BLPF and TLPF? What are the differences between the two waveforms? ()

- A. The signal stabilizes, and transitions ideally. The two waveforms differ in the time required for the transition edges.

- B. The signal stabilizes, and transitions ideally. The two waveforms are the same in the time required for the transition edges.**
- C. The signal oscillates at the transition points (rising and falling edges) before stabilizing. The two waveforms are the same in the time required for the transition edges.**
- D. The signal oscillates at the transition points (rising and falling edges) before stabilizing. The two waveforms differ in the time required for the transition edges.**

(3) Transition Rate of BLPF and TLPF.

Measure the transition rate of both filters during high-to-low and low-to-high level changes. Record the results in Table 1.2. During measurement, use the mouse to move the 4 yellow cursor lines on Lab 4 tab oscilloscope. The difference ΔX between two vertical cursors and the difference ΔY between two horizontal cursors can be read from the table on the right side of the software, where ΔX is in seconds (s) and ΔY is in volts (V). For this experiment, ignore the filter's fluctuation when measuring the level of transition rate.

Table 1.2 Transition Time of Two Filters

Measurement Range	BLPF(1kHz) / μs	TLPF(1kHz) / μs
Rise (10% to 90%)		
Fall (90% to 10%)		

Question 1.3

When gradually increasing the clock frequency of the Pulse Generator, how do the output waveforms of the BLPF and TLPF change? At approximately what frequency does the output undergo significant distortion or become severely altered? ()

- A. When reaching approximately 1000–1500 Hz, the TLPF output undergoes severe changes or even distortion. When reaching approximately 1500–2000 Hz, the BLPF output undergoes severe changes or even distortion.**
- B. When reaching approximately 1500–2000 Hz, the BLPF output undergoes severe changes or even distortion. When reaching approximately 2500–3000 Hz, the TLPF output undergoes severe changes or even distortion.**
- C. When reaching approximately 1500–2000 Hz, the BLPF output undergoes severe changes or even distortion. When reaching approximately 1500–2000 Hz, the TLPF output undergoes severe changes or even distortion.**
- D. When reaching approximately 1000–1500 Hz, the TLPF output undergoes severe changes or even distortion. When reaching approximately 2500–3000 Hz, the BLPF output undergoes severe changes or even distortion.**

(4) Waveform during High-and-low Level Transition.

Set the Pulse Generator frequency to 250 Hz, duty cycle to 50%, and keep other parameters unchanged. Continue observing the outputs of the BLPF and TLPF on the oscilloscope. Plot the transition process from low to high level (or high to low level) for both BLPF and TLPF on the same graph (Figure 1.9), ensuring units are labeled on the axes.

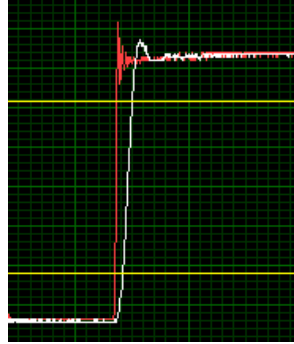


Figure 1.9 Transition process from low level to high level

(5) Transition Rate of Three Filters.

Maintain the Pulse Generator frequency at 250 Hz, duty cycle at 50%, and other parameters unchanged. Measure the transition rate during level changes for three filters (BLPF, TLPF, RC Low-Pass Filter RCLPF). Record the results in Table 1.3. Based on Table 1.3 and ignoring the vibration time of all three filters, please calculate the maximum number of pulse transitions per second that each filter can handle.

Table 1.3 Transition Time of Three Filters

Measurement Range	BLPF(250Hz) / μs	TLPF (250Hz) / μs	RCLPF(250Hz) / μs
Rise (1% to 99%)			
Fall (99% to 1%)			

3. Characteristics of Sinusoidal Signals

In the virtual instrument platform interface, please select the Function Generator, and its parameter settings are shown in Figure 1.10. Connect the wires as shown in Figure 1.11. Channel CH0 of the oscilloscope is connected to the output port of BLPF. Channel CH1 is connected to the output port of TLPF. Open the SIGEx program, and you can observe the output waveforms through Lab 3 tab.

After clicking the "Run" button, gradually increase the frequency from 250 Hz to 10 kHz and observe its effect on the output signal amplitude. Record the peak-to-peak voltage of three filter outputs in different frequencies in Table 1.4, and plot the graph using a logarithmic scale (Figure 1.12). Note: The data in the figure is for illustration only and has no reference value.

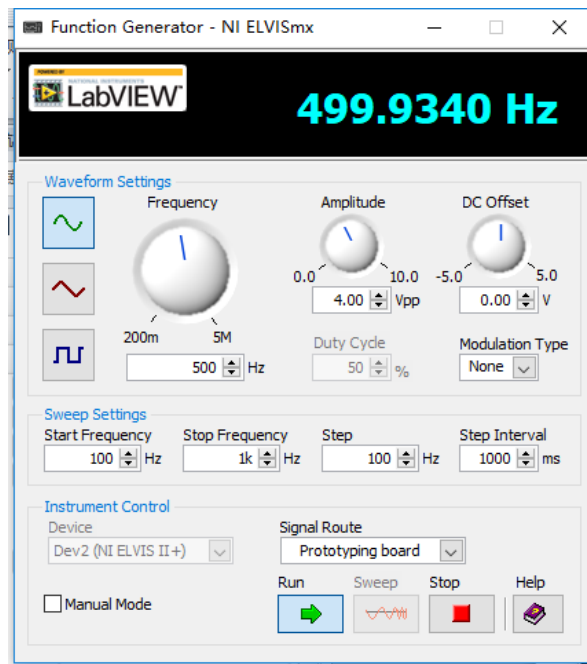


Figure 1.10 Function generator parameter settings

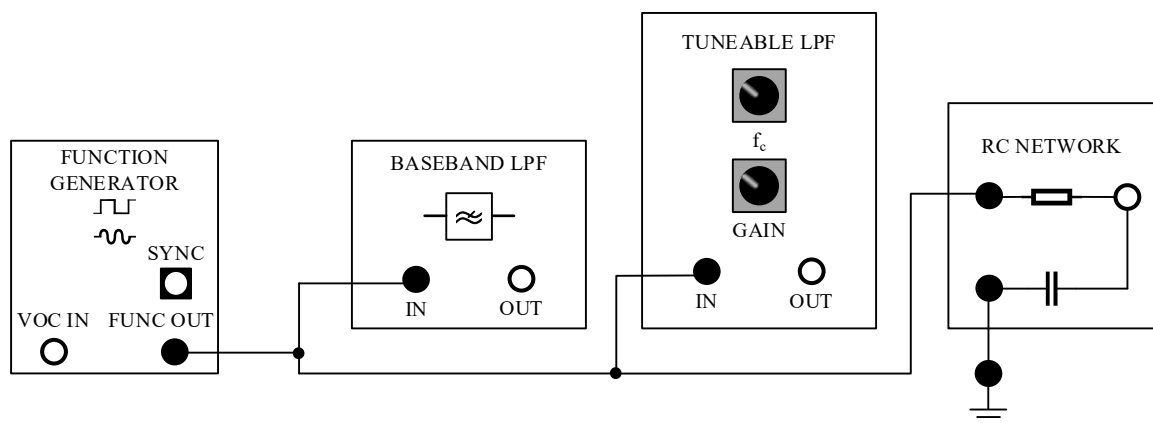


Figure 1.11 Wiring diagram of sine wave passing through filters

Question 1.4

Based on the measurement data of sinusoidal signal characteristics, what are the upper frequency limits where BLPF and TLPF can work normally? ()

- A. 1 kHz, 1 kHz.
- B. 5 kHz, 1 kHz.
- C. 1 kHz, 5 kHz.
- D. 5 kHz, 5 kHz.

Table 1.4 Relationship between filters and frequencies

Frequency (kHz)	BLPF (V _{pp})	TLPF (V _{pp})	RCPF (V _{pp})
0.25			
0.5			
0.8			
1			
1.5			
2			
2.5			
3			
4			
5			
8			
10			

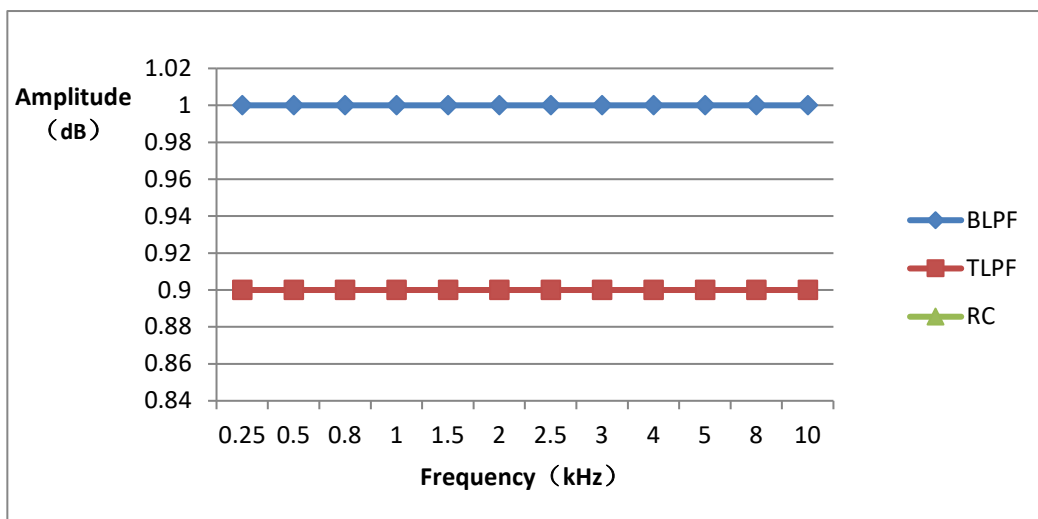


Figure 1.12 Relationship between logarithm of amplitude and frequency

4. Limiting (Optional)

Set the switches on the Limiter (LIMITER) unit to A=ON, B=OFF. Connect the circuit according to Figure 1.13. Use the oscilloscope to observe the input and output of the limiter. Set the Function Generator parameters as follows: sine wave frequency = 1.2 kHz, $V_{pp} = 4$ V.

Change the input signal amplitude using the AMPLITUDE control on the Function Generator and observe the output. Record the phenomena and plot a graph showing the relationship between input voltage peak-to-peak (V_{pp}) and output voltage peak-to-peak.

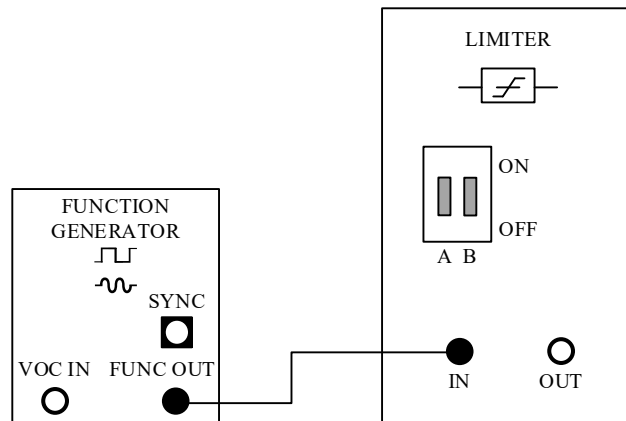


Figure 1.8 Limiter Wiring

Question 1.5

What changes occur to the output of the limiter when the amplitude of the signal is altered through the amplitude control on the signal generator?

VI. Precautions

(1) Perform checks according to the experimental preparation requirements before starting the experiment.

(2) If the instrument loses power unexpectedly during the experiment, restart the NIELVISmx Instrument Launcher. On the SIGEx panel, click the "Stop" button first, then click the "Run" button.

(3) During the experiment, open the cursor legend on the oscilloscope display and create free cursors to assist with readings.

(4) After completing all experiments, turn off the power, disconnect all wires, tidy the workbench, and shut down the computer before leaving the laboratory. Comply with all laboratory rules and regulations.

(5) If the experimental instruments remain unchanged, they will not be introduced separately in subsequent experiments.

VII. Lab Report Requirements

(1) Complete the experiment content independently and record the results honestly.

(2) Answer the experimental questions.